CO2 - AT2 - DEBUGGING TASK

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1. **A clustering algorithm like K-means produces inconsistent cluster assignments for similar datasets. Analyze possible reasons such as poor initialization, incorrect number of clusters, or sensitivity to outliers. Propose debugging approaches and optimization strategies like K-means++, normalization, and silhouette analysis.**

K-Means may produce inconsistent cluster assignments because of several reasons.

Possible Reasons

- **Poor Initialization:** Random selection of initial centroids may lead to different clustering results.
- **Incorrect Number of Clusters (K):** Choosing an inappropriate value of K results in poor cluster formation.
- **Sensitivity to Outliers:** Outliers can significantly affect centroid positions and reduce clustering quality.
- **Different Feature Scales:** Features with larger values dominate distance calculations.

Debugging Approaches

- Run K-Means multiple times and compare the results.
- Visualize the clusters to identify incorrect grouping.
- Detect and remove outliers before clustering.
- Evaluate clustering quality using metrics such as the **Silhouette Score**.

Optimization Strategies

- Use **K-Means++** initialization to select better initial centroids.
- Normalize or standardize features before clustering.
- Determine the optimal value of K using the **Elbow Method** and **Silhouette Analysis**.
- Remove noise and outliers to improve cluster stability.

2. **A support vector machine (SVM) model underperforms on a nonlinear dataset. Identify issues such as incorrect kernel choice, improper parameter tuning, or scaling problems. Suggest debugging techniques and optimization methods including kernel selection, hyperparameter tuning, and feature scaling.**

An SVM model may underperform on a nonlinear dataset due to several factors.

Possible Issues

- Incorrect Kernel Choice: A linear kernel cannot effectively classify nonlinear data.
- Improper Hyperparameter Tuning: Incorrect values of C and Gamma reduce model performance.
- Feature Scaling Problems: SVM is sensitive to the scale of input features.

Debugging Techniques

- Visualize the dataset to understand its distribution.

- Test different kernel functions such as Linear, Polynomial, and RBF.
- Evaluate model performance using cross-validation and confusion matrices.
- Check whether all features are properly scaled.

Optimization Methods

- Select an appropriate kernel, such as the RBF kernel, for nonlinear datasets.
- Perform hyperparameter tuning using Grid Search or Random Search.
- Apply feature scaling using Standardization or Normalization.
- Use cross-validation to identify the best parameter combination.

3. A natural language processing (NLP) model using recurrent neural networks (RNNs) suffers from poor long-term dependency learning. Examine causes such as vanishing gradients and limited memory capacity. Describe debugging steps and propose solutions like LSTM/GRU units, attention mechanisms, and improved training strategies.

RNN models often struggle to learn long-term dependencies in sequential data.

Possible Causes

- Vanishing Gradients: Gradients become very small during backpropagation, preventing the model from learning long-term relationships.
- Limited Memory Capacity: Standard RNNs cannot effectively retain information over long sequences.
- Insufficient Training: Poor training strategies or inadequate data may reduce performance.

Debugging Steps

- Monitor training and validation loss to detect learning problems.
- Analyze gradient values to identify vanishing gradient issues.
- Test the model on sequences of different lengths.
- Compare RNN performance with advanced architectures.

Optimization Strategies

- Replace standard RNNs with LSTM or GRU networks, which retain long-term information more effectively.
- Use Attention Mechanisms to focus on important words in the sequence.
- Apply improved training techniques such as gradient clipping, proper learning rate scheduling, and regularization.
- Increase training data and use pretrained word embeddings to improve model performance.